EMESYS Portal - Complete Working Guide

Admin + Teacher + Student + Parent workflows for demo/presentation

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Purpose

This document explains how the full portal works in practical sequence: setup by Admin, day-to-day operations by Teacher, learning actions by Student, and monitoring by Parent. It is prepared for senior-level walkthrough and handover.

Quick System Flow

Admin Setup -> Teacher Operations -> Student Actions -> Parent Visibility -> Notifications & Analytics

1. Role Matrix and Responsibilities

Role	Primary Responsibility	Where it is controlled
Admin / Principal	Create sections, assign teachers, create section subjects	Django Admin + restricted teacher subject pages
Teacher / Class Teacher	Assignments, exams, attendance, notes, student management	Teacher dashboard pages
Student	View subjects, attempt assignments/exams, raise queries, track progress	Student dashboard pages
Parent	Monitor child performance, assignments, attendance, messages	Parent dashboard pages

Access control is role-driven and section-aware. Teachers and students only see data mapped to their sections/subjects. Parents see only linked child data.

2. Admin / Principal Setup Workflow

2.1 Create base masters

- Create users with correct role: Admin/Principal/Teacher/Class Teacher/Student/Parent.
- Create TeacherProfile, StudentProfile, ParentProfile records.
- Create ParentStudentRelation to link each parent with one or more children.

2.2 Create sections (single or bulk)

- Section master stores class level and section (example: UKG-A, 5-B).
- Bulk section creation is supported in admin (example input: A,B,C for same class level).
- Class teacher can be set while creating/editing section.

2.3 Assign teacher to section and subjects

- TeacherSectionAssignment maps teacher to section (with option Is Class Teacher).
- Admin can map one teacher to multiple sections in one action.
- Optional subject mapping in same flow creates SubjectTeacherAssignment records.
- Validation ensures subject class level matches section class level.
- Validation also ensures one section-subject has only one active subject teacher.

2.4 Create section-wise subjects

- Subject creation page is restricted to Admin/Principal only.
- Subject is created with section context and assigned subject teacher.
- Duplicate section+subject assignment is blocked with clear message.

3. Teacher Portal - Daily Working

3.1 Dashboard and class data scope

- Teacher dashboard metrics are dynamic (students, active assignments, pending queries, upcoming tests).
- All views are filtered by accessible sections and student mappings.

3.2 Assign homework / class test

- Teacher opens Assign Questions page and selects a valid section.
- Teacher enters description, marks, due date, optional file/LaTeX.
- System creates Question record mapped to selected section.
- Section-specific students get notifications (no cross-section broadcast).
- Linked parents also receive assignment notification.

3.3 Create exam and questions

- Teacher creates exam with title, due date, total marks, timer/duration (optional), section, subject, chapter.
- Multiple exam questions can be attached with marks and files.
- Exam is visible only to students of target section.

3.4 Upload notes and subject resources

- Teacher Notes: title, message, optional file, section and subject scope.
- Subject Resources: Books, Notes, Video, Quiz, Worksheet under subject.
- Only assigned subject teacher can upload for that section-subject.
- Students see only published resources for their section/class visibility.

3.5 Attendance, remarks, and communication

- Attendance can be marked by date and section (Present/Absent/Late).
- Upsert logic prevents duplicate attendance row for same student/date.
- Teacher reviews submissions and gives marks + remarks.
- Teacher can respond to student queries and post class messages/announcements.

4. Student Portal - Learning Journey

4.1 What student sees on login

- Dashboard cards: pending assignments, queries, upcoming exams, graded count.
- Subject cards with assigned teacher and resource counters (book/note/video/quiz/worksheet).
- Recent subject notes and notification stream.

4.2 Assignment and exam flow

- Student opens pending assignment list (section-filtered).
- Student submits text/file answer before due date.
- Teacher evaluates and publishes marks/remarks.
- Student can attempt section-specific exams and submit per-question answers.
- Submission and grading events trigger notifications.

4.3 Notes, resources, and performance

- Class Notes page shows teacher notes for student section.
- My Subjects page shows resource availability per subject.

- Attendance and performance pages show student-level analytics.
- Practice with AI and chapter learning pages support extra revision.

5. Parent Portal - Monitoring and Feedback

5.1 Child-based dashboard

- Parent selects linked child (if multiple children).
- Dashboard shows attendance %, pending assignments, upcoming exams, and recent notifications.
- All cards are scoped to selected child section.

5.2 Assignments and parent-friendly review

- Parent can open assignments with marks and teacher remarks.
- AI tools provide simplified and parent-friendly feedback language.
- Feedback format can be made actionable (what went well, support area, this week action).

5.3 Messages, notes, holidays, exam portion

- Parent receives class messages/announcements relevant to child section.
- Class notes and upcoming exam portion are section-filtered.
- Holiday calendar combines global + class + section scope.

6. End-to-End Scenarios (Presentation Friendly)

Scenario	Action Chain	Visible To
Section Subject Setup	Admin creates section -> assigns teacher -> creates subject -> maps subject to section	Teacher, Student sees subject card
Assignment Lifecycle	Teacher assigns question to section -> student submits -> teacher marks -> notification	Teacher, Student, Parent
Exam Lifecycle	Teacher creates exam/questions -> student attempts -> teacher evaluates	Teacher, Student, Parent
Notes Distribution	Teacher uploads notes/resource in section-subject -> student/parent sees in class notes view	Teacher, Student, Parent
Attendance Lifecycle	Teacher marks attendance -> student attendance page updates -> parent dashboard reflects summary	Teacher, Student, Parent

7. Key Controls and Validations

- Strict section-based targeting for assignment/exam notifications.
- SubjectTeacherAssignment checks class-level compatibility.
- Single section-subject cannot have multiple active subject teachers.
- Attendance unique constraint: one record per student per day.
- Role-based access methods ensure users only access allowed records.

8. 10-Minute Demo Script for Senior Presentation

- Minute 1-2: Show Admin setup (Section creation + Teacher assignment).
- Minute 3-4: Show Admin/Principal creating section subject and assigning subject teacher.
- Minute 5-6: Login as Teacher and assign one homework + one exam to a section.
- Minute 7: Login as Student and show assignment/exam visibility and submission.
- Minute 8: Teacher checks submission and gives marks + remark.
- Minute 9: Login as Parent and show performance card + parent-friendly feedback.
- Minute 10: Show notification trail and section-specific visibility proof.

Appendix: Important URL Prefixes

Admin: /admin/ | Teacher: /portal/dashboard/teacher/... | Student: /portal/dashboard/student/... | Parent: /portal/dashboard/parent/...

Prepared for internal review and operational handover. This guide is based on implemented Django views, models, access-control rules, and signal-driven notifications.